

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q19-1169

Torrington Terminal 138 kV
August 22, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1169. The Interconnection Customer proposes a 115.1 MW solar photovoltaic facility interconnecting at the Torrington Terminal 138 kV substation in the Virginia area. The request was received on March 17, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 382 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$769,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Declo - New London Substation 138kV: 106.4% loading (425 MVA) following the outage of the Bath County - Sherburne 345kV L; DFAX = 5.1%.

Lowmoor - Longview2 345kV Line: 103.1% loading (1,268 MVA) following the outage of the Longview2 - Bath County 345kV; DFAX = 10.3%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
New London - Declo 138kV	Reconductor line	\$13,835,000	3% (\$416,000)
Lowmoor - Longview2 345kV	Reconductor line	\$35,929,000	1% (\$353,000)
	Total allocated to project		\$769,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.